Length Rigidity and the Nowhere-Continuous-Inverse Obstruction for Scottish Book Problem 155

Partial-result packet for arXiv:2308.03339

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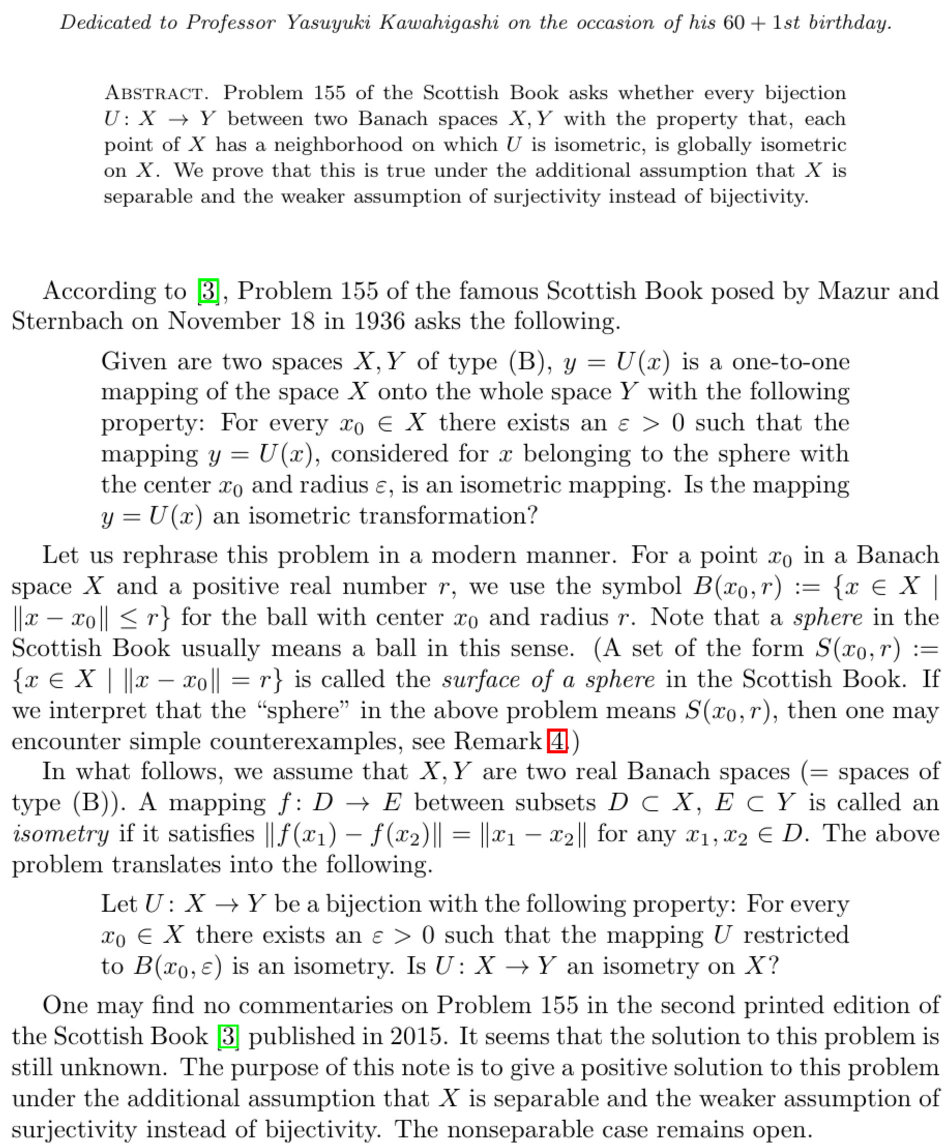
Status: substantial partial result, likely valid; full problem open. A locally distance-preserving surjection between Banach spaces is globally nonexpansive and preserves path length. Continuity of the inverse at one point forces a global affine isometry. Thus every counterexample must have a nowhere-continuous inverse and closed nowhere-dense local sheets. An exact locally isometric but globally contracting self-embedding of ℓ_∞ shows why surjectivity is the remaining essential obstacle.

1 Source question

Mori [1] studies Scottish Book Problem 155. In modern notation it asks: if $U : X \to Y$ is a bijection between real Banach spaces and for every $x \in X$ there is $r_x > 0$ such that $U|_{B(x,r_x)}$ preserves distances, must U preserve all distances? The source proves this when X is separable and explicitly leaves the nonseparable case open.



Dedicated to Professor Yasuyuki Kawahigashi on the occasion of his $60+1$st birthday.

Abstract. Problem 155 of the Scottish Book asks whether every bijection $U\colon X \to Y$ between two Banach spaces X, Y with the property that, each point of X has a neighborhood on which U is isometric, is globally isometric on X. We prove that this is true under the additional assumption that X is separable and the weaker assumption of surjectivity instead of bijectivity.

According to [3], Problem 155 of the famous Scottish Book posed by Mazur and Sternbach on November 18 in 1936 asks the following.

> Given are two spaces X, Y of type (B), $y = U(x)$ is a one-to-one mapping of the space X onto the whole space Y with the following property: For every $x_0 \in X$ there exists an $\varepsilon > 0$ such that the mapping $y = U(x)$, considered for x belonging to the sphere with the center x_0 and radius ε, is an isometric mapping. Is the mapping $y = U(x)$ an isometric transformation?

Let us rephrase this problem in a modern manner. For a point x_0 in a Banach space X and a positive real number r, we use the symbol $B(x_0, r) := \{x \in X \mid \|x - x_0\| \le r\}$ for the ball with center x_0 and radius r. Note that a *sphere* in the Scottish Book usually means a ball in this sense. (A set of the form $S(x_0, r) := \{x \in X \mid \|x - x_0\| = r\}$ is called the *surface of a sphere* in the Scottish Book. If we interpret that the "sphere" in the above problem means $S(x_0, r)$, then one may encounter simple counterexamples, see Remark 4.)

In what follows, we assume that X, Y are two real Banach spaces (= spaces of type (B)). A mapping $f\colon D \to E$ between subsets $D \subset X$, $E \subset Y$ is called an *isometry* if it satisfies $\|f(x_1) - f(x_2)\| = \|x_1 - x_2\|$ for any $x_1, x_2 \in D$. The above problem translates into the following.

> Let $U\colon X \to Y$ be a bijection with the following property: For every $x_0 \in X$ there exists an $\varepsilon > 0$ such that the mapping U restricted to $B(x_0, \varepsilon)$ is an isometry. Is $U\colon X \to Y$ an isometry on X?

One may find no commentaries on Problem 155 in the second printed edition of the Scottish Book [3] published in 2015. It seems that the solution to this problem is still unknown. The purpose of this note is to give a positive solution to this problem under the additional assumption that X is separable and the weaker assumption of surjectivity instead of bijectivity. The nonseparable case remains open.

Figure 1: The source question and its nonseparable context (PDF page 1).

Basso [2] proves the arbitrary-density case under the stronger assumption that every local restriction maps onto an *open* target neighborhood. That hypothesis makes U a covering/local-isometry map; it is not automatic in the question above.

2 Proof intuition

The straight segment from x to y is compact, so finitely many of the isometric neighborhoods control its image. Its image has the same path length, which immediately yields the one-sided global estimate $\|U(x) - U(y)\| \le \|x - y\|$. The missing inequality is precisely a lifting problem for target paths.

One open local image is enough to remove that obstruction: Mankiewicz's extension theorem supplies a global affine isometry, and agreement propagates through connectedness. Hence one continuity point of U^{-1} settles the whole problem. If there is no such point, all complete local images must be nowhere dense. The final construction shows that thin local images can indeed fold distant points close together, even in ℓ_∞; it fails only the source's surjectivity requirement.

3 Length rigidity and a one-point criterion

The next statement allows surjectivity without injectivity until the inverse is mentioned.

Theorem 1. *Let X, Y be real Banach spaces and let $U : X \to Y$ be a surjection such that every $x \in X$ has a radius $r_x > 0$ for which $U|_{B(x,r_x)}$ is an isometric embedding. Then:*

1. *U preserves the length of every rectifiable path and is globally 1-Lipschitz;*
2. *if some local image $U(B(x, r))$, on which U is isometric, has nonempty interior, then U is a global affine surjective isometry;*
3. *if U is bijective and U^{-1} is continuous at one point of Y, then U is a global affine surjective isometry.*

Proof. Let $\gamma : [0, 1] \to X$ be a rectifiable path. Compactness of its image and the local hypothesis give a subdivision fine enough that each subarc lies in one isometric ball. Refining any partition in this way shows that the sums of successive distances used to define the lengths of γ and $U \circ \gamma$ agree. Taking suprema gives $L(U \circ \gamma) = L(\gamma)$. Applying this to the linear segment from x to y yields

$$\|U(x) - U(y)\| \leq L(U \circ \gamma) = L(\gamma) = \|x - y\|.$$

For (2), Mori's Lemma 2, which is an application of Mankiewicz's extension theorem, says that the local restriction extends uniquely to an affine surjective isometry $A : X \to Y$. Let D be the set of points near which $U = A$. It is nonempty and open. If $D \neq X$, choose a boundary point z. On a sufficiently small isometric ball at z, the maps agree on a nonempty open subset coming from D. Consequently the image of that local ball has nonempty interior; its unique affine extension must be A. Hence $U = A$ in a neighborhood of z, a contradiction. Connectedness of X gives $D = X$.

For (3), write $y = U(x)$ for a continuity point of U^{-1} and choose a local isometric ball $B(x, r)$. Continuity at y provides $\delta > 0$ with

$$U^{-1}(B(y, \delta)) \subset B(x, r).$$

Surjectivity and injectivity therefore give $B(y, \delta) \subset U(B(x, r))$. Apply (2). □

Corollary 2. *Under the hypotheses of Theorem 1, if U is bijective and $U^{-1} : Y \to X$ is of Baire class one, then U is a global affine surjective isometry.*

Proof. A Baire-one map from the completely metrizable space Y to a metric space has a point of continuity. Use Theorem 1(3). □

4 The category obstruction

Let $\mathrm{cov}_{\mathrm{nd}}(Y)$ be the least cardinality of a cover of Y by closed nowhere-dense sets. Let $\mathrm{dens}(X)$ denote the density character of X.

Proposition 3. *If*

$$\mathrm{dens}(X) < \mathrm{cov}_{\mathrm{nd}}(Y),$$

then every map in Theorem 1 is a global affine surjective isometry.

Proof. For each x, first replace its local radius r_x by the open ball $B^\circ(x, r_x/4)$. These smaller balls cover X. A metric space of density κ has Lindelöf number at most κ, so select a subcover of cardinal at most $\mathrm{dens}(X)$. Replace each selected $B^\circ(x, r_x/4)$ by $B(x, r_x/2)$; the closed balls still cover X and lie inside isometric neighborhoods. The image of each is closed because it is isometric to a complete space. If all these images were nowhere dense, their union could not cover Y by the displayed cardinal inequality. Therefore one has interior, and Theorem 1(2) applies. $\square$

For separable X, Proposition 3 reduces to the ordinary Baire-category argument in the source. Its point here is structural: in any genuine nonseparable counterexample, every complete local sheet is closed nowhere dense, and enough such sheets must cover Y.

5 A sharp near-counterexample on the bounded-sequence space

Define the one-periodic tent wave

$$\tau(t) = \mathrm{dist}(t, \mathbb{Z})$$

and

$$F(t) = \big(\tau(t), \tau(t + 1/4), \tfrac{1}{10}\arctan t\big) \in \ell^3_\infty.$$

Proposition 4. *There is an injective self-map $T : \ell_\infty \to \ell_\infty$ whose restriction to every ball of radius $1/16$ is isometric, but T is not a global isometric embedding. In particular, the surjectivity hypothesis in Scottish Book Problem 155 cannot be discarded.*

Proof. The corner sets of $\tau(t)$ and $\tau(t + 1/4)$ are respectively $\frac{1}{2}\mathbb{Z}$ and $\frac{1}{2}\mathbb{Z} - \frac{1}{4}$. For every $a \in \mathbb{R}$, one of these two sets is at distance at least $1/8$ from a. The corresponding tent coordinate is affine with slope 1 or -1 throughout the radius-$1/16$ neighborhood of a. All three coordinates of F are 1-Lipschitz, while that chosen coordinate witnesses equality. Thus

$$\|F(s) - F(t)\|_\infty = |s - t| \quad (s, t \in B(a, 1/16)). \tag{1}$$

The last coordinate makes F injective. On the other hand, for integers $N > 0$,

$$\|F(N) - F(0)\|_\infty = \tfrac{1}{10}\arctan N < N = |N - 0|. \tag{2}$$

Apply F coordinatewise:

$$\mathcal{F}((x_i)_{i\in\mathbb{N}}) = (F_j(x_i))_{(i,j)\in\mathbb{N}\times\{1,2,3\}}.$$

Equation (1), uniformly in i, shows that $\mathcal{F}$ is isometric on every radius-$1/16$ ball of ℓ_∞; it is injective, and (2) applied to constant sequences shows global contraction. Finally reindex $\mathbb{N} \times \{1, 2, 3\}$ by $\mathbb{N}$ to obtain the claimed self-map T. $\square$

The range of T satisfies a coordinatewise curve constraint and is not all of ℓ_∞. Attempts to absorb this constraint by shifts or triangular shears either preserved a non-surjective range condition or destroyed local sup-distance equality. This is the remaining gap between the construction and a counterexample to the source problem.

6 Verification, scope, and novelty

The accompanying script checked 20,000 pseudorandom pairs in common radius-$1/16$ neighborhoods and the global contraction in (2). This is only a transcription audit; Proposition 4 is proved exactly above.

Eight focused upgrade attempts covered length lifting, convex bicombings, midpoint characterizations, inverse continuity, generalized category, Baire-one regularity, the tent-wave construction, and surjective absorption. The full bijective nonseparable problem remains open because no argument forces a target segment to lift and no locally isometric thin embedding was made surjective.

On 12 August 2026, bounded searches of the four run indexes and current literature used the source id, Problem 155, Mazur–Sternbach, the nonseparable case, and later citations. They found no full resolution. Basso's result uses the strictly stronger open-image hypothesis. Novelty confidence for the elementary partial criteria is moderate-low; validity confidence is high.

Human-review recommendation. Review as a partial result. The most useful checks are the closed-ball subcover sentence in Proposition 3, the one-point inverse continuity reduction, and the uniform radius in Proposition 4.

References

[1] M. Mori, *On the Scottish Book Problem 155 by Mazur and Sternbach*, arXiv:2308.03339, 2023; C. R. Math. 362 (2024), 813–816.

[2] G. Basso, *On a problem of Mazur and Sternbach*, arXiv:2308.08400v2, 2023.